Worksheet 1.5 — Exponents & Radicals

Mathematics: Analysis & Approaches SL (IB) · Unit 1: Number and Algebra

The Big Ideas

The laws of exponents and radicals let you simplify expressions and solve equations where the variable sits in an exponent.

- $a^m a^n = a^{m+n}$, $\dfrac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$, $a^{-n} = \dfrac{1}{a^n}$.
- $a^{1/n} = \sqrt[n]{a}$, $a^{m/n} = (\sqrt[n]{a})^m$.
- Matching-base equations: write both sides with the same base, then equate exponents.

Worked Examples

Example 1 — Simplify

Simplify $\dfrac{(x^3y^{-2})^2}{x^{-1}y^4}$.

Solution. Numerator: x^6y^{-4}. Divide: $x^6y^{-4} \div x^{-1}y^4 = x^{6-(-1)}y^{-4-4} = x^7y^{-8}$.

Conclusion: $\dfrac{x^7}{y^8}$.

Example 2 — Simplify radicals

Simplify $\sqrt{75} - \sqrt{12} + \sqrt{27}$.

Solution. $\sqrt{75} = 5\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$. Combine: $5\sqrt{3} - 2\sqrt{3} + 3\sqrt{3}$.

Conclusion: $6\sqrt{3}$.

Example 3 — Solve an exponential equation

Solve $4^{2x-1} = 64$.

Solution. Write $64 = 4^3$: $2x - 1 = 3 \Rightarrow x = 2$.

Conclusion: $x = 2$.

Example 4 — Rationalize

Rationalize $\dfrac{6}{\sqrt{7}+2}$.

Solution. Multiply by the conjugate: $\dfrac{6(\sqrt{7}-2)}{(\sqrt{7}+2)(\sqrt{7}-2)} = \dfrac{6(\sqrt{7}-2)}{7-4} = \dfrac{6(\sqrt{7}-2)}{3}$.

Conclusion: $2(\sqrt{7}-2) = 2\sqrt{7} - 4$.

Example 5 — Simplify a fractional exponent

Simplify $\left(\dfrac{16x^8}{81}\right)^{3/4}$.

Solution. Apply the exponent to numerator and denominator separately: $\dfrac{(16)^{3/4}(x^8)^{3/4}}{(81)^{3/4}} = \dfrac{8x^6}{27}$.

Conclusion: $\dfrac{8x^6}{27}$.

Example 6 — Solve with unmatched-looking bases

Solve $8^{x+1} = 32^{x-1}$ by writing both sides with base 2.

Solution. $8 = 2^3$, $32 = 2^5$: $2^{3x+3} = 2^{5x-5} \Rightarrow 3x + 3 = 5x - 5$.

Conclusion: $8 = 2x \Rightarrow x = 4$.

Example 7 — Rationalize with a binomial denominator

Rationalize $\dfrac{2+\sqrt{3}}{2-\sqrt{3}}$.

Solution. Multiply by the conjugate: $\dfrac{(2+\sqrt{3})^2}{(2-\sqrt{3})(2+\sqrt{3})} = \dfrac{4+4\sqrt{3}+3}{4-3}$.

Conclusion: $7 + 4\sqrt{3}$.

Example 8 — Simplify nested radicals

Simplify $\sqrt{2}\cdot\sqrt{50} + \sqrt{8}\cdot\sqrt{2}$.

Solution. $\sqrt{2}\cdot\sqrt{50} = \sqrt{100} = 10$; $\sqrt{8}\cdot\sqrt{2} = \sqrt{16} = 4$.

Conclusion: $10 + 4 = 14$.

Example 9 — Solve a substitution-style exponential equation

Solve $3^{2x} - 10(3^x) + 9 = 0$ (let $u = 3^x$).

Solution. $u^2 - 10u + 9 = 0 \Rightarrow (u-1)(u-9) = 0 \Rightarrow u = 1$ or $u = 9$.

Conclusion: $3^x = 1 \Rightarrow x = 0$; $3^x = 9 \Rightarrow x = 2$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Simplify $\dfrac{x^5y^{-1}}{x^{-2}y^3}$.

Question 2

Simplify $\sqrt{50}+\sqrt{8}-\sqrt{18}$.

Question 3

Solve $2^{3x+1}=32$.

Question 4

Simplify $32^{3/5}$.

Question 5

Rationalize $\dfrac{4}{\sqrt{5}-1}$.

Question 6

Solve $9^{x-1} = 27^{2x}$ by writing both sides with base 3.

Question 7

Simplify $\left(\frac{8x^6}{27}\right)^{1/3}$.

Question 8

Simplify $\frac{\sqrt{45}}{\sqrt{5}} + \sqrt{2} \cdot \sqrt{8}$.

Question 9

Rationalize $\frac{3-\sqrt{2}}{3+\sqrt{2}}$.

Question 10

Simplify $\left(\frac{25x^4}{9}\right)^{3/2}$.

Question 11 — Challenge

Solve $2^{2x} - 5(2^x) + 4 = 0$ (let $u = 2^x$, factor as a quadratic in u).

Question 12 — Challenge

Solve $16^{x+1} = 64^{x-2}$ by writing both sides with base 2.

Question 13 — Challenge

Solve $5^{2x} - 6(5^x) + 5 = 0$ (let $u = 5^x$).

Question Recap & Answer Key

Question Recap

1. Simplify $\dfrac{x^5y^{-1}}{x^{-2}y^3}$.
2. Simplify $\sqrt{50}+\sqrt{8}-\sqrt{18}$.
3. Solve $2^{3x+1}=32$.
4. Simplify $32^{3/5}$.
5. Rationalize $\dfrac{4}{\sqrt{5}-1}$.
6. Solve $9^{x-1}=27^{2x}$ by writing both sides with base 3.
7. Simplify $\left(\dfrac{8x^6}{27}\right)^{1/3}$.
8. Simplify $\dfrac{\sqrt{45}}{\sqrt{5}}+\sqrt{2}\cdot\sqrt{8}$.
9. Rationalize $\dfrac{3-\sqrt{2}}{3+\sqrt{2}}$.
10. Simplify $\left(\dfrac{25x^4}{9}\right)^{3/2}$.
11. Solve $2^{2x}-5(2^x)+4=0$ (let $u=2^x$, factor as a quadratic in u).
12. Solve $16^{x+1}=64^{x-2}$ by writing both sides with base 2.
13. Solve $5^{2x}-6(5^x)+5=0$ (let $u=5^x$).

✓ Answer Key

1. $x^7y^{-4}=\dfrac{x^7}{y^4}$
2. $5\sqrt{2}+2\sqrt{2}-3\sqrt{2}=4\sqrt{2}$
3. $3x+1=5\Rightarrow x=\frac{4}{3}$
4. $32^{3/5}=(2^5)^{3/5}=2^3=8$
5. $\dfrac{4(\sqrt{5}+1)}{4}=\sqrt{5}+1$
6. $3^{2x-2}=3^{6x}\Rightarrow 2x-2=6x\Rightarrow x=-\frac{1}{2}$
7. $\dfrac{2x^2}{3}$
8. $\sqrt{9}+\sqrt{16}=3+4=7$
9. $\dfrac{(3-\sqrt{2})^2}{9-2}=\dfrac{9-6\sqrt{2}+2}{7}=\dfrac{11-6\sqrt{2}}{7}$
10. $\dfrac{125x^6}{27}$
11. $u^2-5u+4=0\Rightarrow(u-1)(u-4)=0\Rightarrow u=1$ or $u=4$; $2^x=1\Rightarrow x=0$; $2^x=4\Rightarrow x=2$
12. $2^{4x+4}=2^{6x-12}\Rightarrow 4x+4=6x-12\Rightarrow x=8$
13. $u^2-6u+5=0\Rightarrow(u-1)(u-5)=0\Rightarrow u=1$ or $u=5$; $5^x=1\Rightarrow x=0$; $5^x=5\Rightarrow x=1$